

U19BM603- Biomechanics

UNIT II

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CONSTITUTIVE EQUATIONS

CONSTITUTIVE EQUATIONS

- The properties of materials are specified by constitutive equations. A wide variety of materials exist.
- Three simple, idealized stress-strain relationships, namely, the **nonviscous fluid**, **the Newtonian viscous fluid**, and **the Hookean elastic solid**, give a good description of the mechanical properties of many materials around us.
- Within certain limits of strain and strain rate, air, water, and many engineering structural materials can be described by these idealized equations.

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- A constitutive equation describes a physical property of a material.
- Hence it must be independent of any particular set of coordinates of reference with respect to which the components of various physical quantities are resolved.
- Hence a constitutive equation must be a tensor equation: Every term in the equation must be a tensor of the same rank.

The Nonviscous Fluid

- A nonviscous fluid is one for which the stress tensor is of the form
$$\sigma_{ij} = -p\delta_{ij},$$
- where δ_{ij} is the Kronecker delta (function of two variables, usually just non-negative integers), which has the value 1 if $i = j$, and zero if $i \neq j$, and p is a scalar called pressure
- The pressure p in an ideal gas $\frac{p}{\rho} = RT$, to the density and temperature T by the equation of state
 - where R is the gas constant.

The Newtonian Viscous Fluid

- A Newtonian viscous fluid is a fluid for which the **shear stress is linearly proportional to the strain rate**.
- For a Newtonian fluid the stress-strain relationship is specified by the equation,

$$\sigma_{ij} = -p\delta_{ij} + \mathcal{D}_{ijkl}V_{kl}, \quad (1)$$

Where σ_{ij} is the **stress tensor**, V_{kl} is the **strain rate tensor**, $\mathcal{D}_{ijkl}$ is a tensor of **viscosity coefficients of the fluid**, and p is the **static pressure**.

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- The term $-p\delta_{ij}$ represents the state of stress possible in a fluid at rest (when $V_{kl} = 0$).
- The static pressure p is assumed to **depend on the density and temperature** of the fluid according to an equation of state.
- For a Newtonian fluid we assume that the elements of the tensor may **depend on the temperature but not on the stress or the rate of deformation.**

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- If a tensor has the same array of components when the frame of reference is rotated or reflected, then it is said to be an isotropic tensor.
- It can be shown that, in a three-dimensional Euclidean space, there are two independent isotropic tensors, namely, $\delta_{ij}\delta_{kl}$ and $\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$.
- If the tensor $\mathcal{D}_{ijkl}$ is isotropic, then it can be expressed in terms of two independent constants λ and μ ,
- $$\mathcal{D}_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}). \quad \square (2)$$

(Contd..)

- On substituting Eq. (2) into Eq. (1), we obtain an isotropic constitutive equation:

$$\sigma_{ij} = -p\delta_{ij} + \lambda V_{kk}\delta_{ij} + 2\mu V_{ij}. \quad (3)$$

- If it is assumed that the mean normal stress $\frac{1}{3}\sigma_{kk}$ is independent of the rate of dilation V_{kk} then we must set, $3\lambda + 2\mu = 0$;

- Then, the constitutive equation becomes $\sigma_{ij} = -p\delta_{ij} + 2\mu V_{ij} - \frac{2}{3}\mu V_{kk}\delta_{ij}.$

(Contd..)

- A fluid that obeys Eq. is called a Stokes fluid, for which one material constant μ , the coefficient of viscosity, suffices to define its property.
- If a fluid is incompressible, then $V_{kk} = 0$, and we have the constitutive equation for an incompressible viscous fluid:

$$\sigma_{ij} = -p\delta_{ij} + 2\mu V_{ij}.$$

- If $\mu = 0$, we obtain the constitutive equation of the nonviscous fluid:

$$\sigma_{ij} = -p\delta_{ij}.$$

The Hookean Elastic Solid

- A Hookean elastic solid is a solid that obeys Hooke's law, which states that the stress tensor is linearly proportional to the strain tensor; i.e.,

$$\sigma_{ij} = C_{ijkl}e_{kl}, \quad (1)$$

- where σ_{ij} is the stress tensor, e_{kl} is the strain tensor, and C_{ijkl} is a tensor of elastic constants, or moduli, which are independent of stress or strain.

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- A great reduction in the number of elastic constants is obtained when the material is isotropic, i.e., when the constitutive equation is isotropic, and the array of elastic constants C_{ijkl} remains unchanged with respect to rotation and reflection of coordinates.
- D_{ijkl} in Newtonian Viscous Fluid, an isotropic material has exactly two independent elastic constants, for which the Hooke's law reads

$$\sigma_{ij} = \lambda e_{\alpha\alpha} \delta_{ij} + 2\mu e_{ij}. \quad (2)$$

(Contd..)

- The constants λ and μ are called the Lamé constants.
- The second constant μ is practically always written as G and identified as the shear modulus.
- Writing out Eq. (2) in extension, and with x, y, z as rectangular cartesian coordinates, we have Hooke's law for an isotropic elastic solid:

$$\begin{aligned}\sigma_{xx} &= \lambda(e_{xx} + e_{yy} + e_{zz}) + 2Ge_{xx}, \\ \sigma_{yy} &= \lambda(e_{xx} + e_{yy} + e_{zz}) + 2Ge_{yy}, \\ \sigma_{zz} &= \lambda(e_{xx} + e_{yy} + e_{zz}) + 2Ge_{zz}, \\ \sigma_{xy} &= 2Ge_{xy}, \quad \sigma_{yz} = 2Ge_{yz}, \quad \sigma_{zx} = 2Ge_{zx}.\end{aligned}\tag{3}$$

- The constants E, ν , and G are related to the Lamé constants λ and G (or μ).
- E is called Young's modulus, ν is called Poisson's ratio, and G is called the modulus of elasticity in shear, or shear modulus.

Stress

- If we want to determine the strength of a tendon, we can test a large or small specimen.
- A large specimen can sustain a large force, a smaller specimen can take only a small force.
- Obviously, the size of the test specimen is incidental; only the force relative to the size is important.
- Thus, we are led to the concept that it is the stress (force per unit cross-sectional area) that is related to the strength of the material.

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- Let the cross-sectional area of a tendon be A , and let the force that acts in the tendon be F .
- The ratio

$$\sigma = F/A$$

- is the stress in the tendon.
- In the International System of Units (SI units), the basic unit of force is the newton (N) and that of length is the meter (m).
- Thus the basic unit of stress is newton per square meter (N/m²) or pascal (Pa, in honor of Pascal).

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- Consider a material continuum B occupying a spatial region V (Fig).
- Imagine a small surface element of area ΔS .
- Let us draw, from a point on ΔS , a unit vector* $\mathbf{v}$ normal to ΔS .
- Then we can distinguish the two sides of ΔS according to the direction of $\mathbf{v}$.
- Let the side to which this normal vector points be called the positive side. Consider the part of material lying on the positive side.

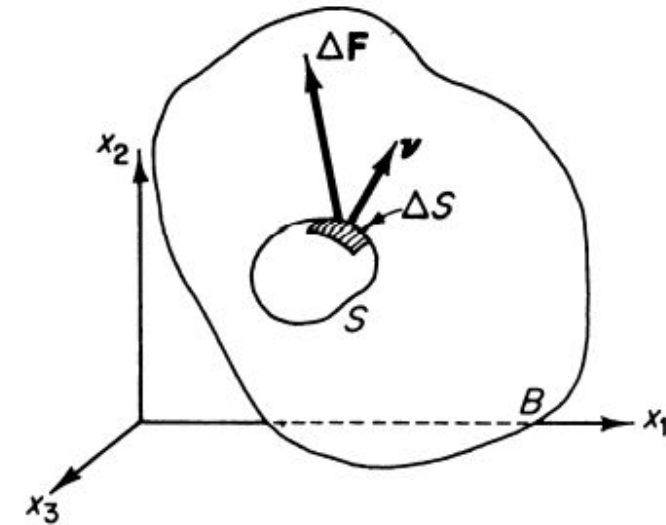


Figure 2.2:1 Stress principle.

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- This part exerts a force ΔF on the other part, which is situated on the negative side of the normal.
- The force ΔF depends on **the location and size of the area** and the orientation of the normal.
- We introduce the assumption that as ΔS tends to zero, the ratio $\Delta F / \Delta S$ tends to a definite limit, dF/dS , and that the moment of the force acting on the surface ΔS about any point within the area vanishes in the limit.
- The limiting vector will be written as $\vec{T} = d\mathbf{F}/dS$,

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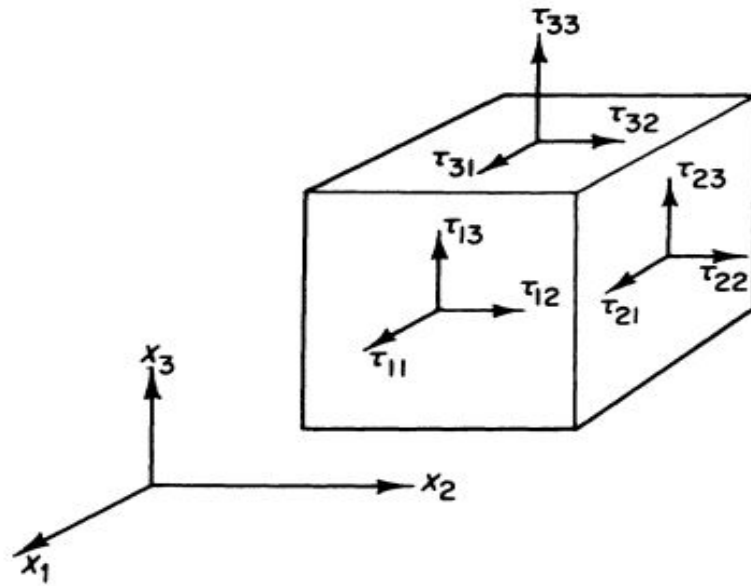


Figure 2.2:2 Notations of stress components.

	Components of Stresses		
	1	2	3
Surface normal to x_1	τ_{11}	τ_{12}	τ_{13}
Surface normal to x_2	τ_{21}	τ_{22}	τ_{23}
Surface normal to x_3	τ_{31}	τ_{32}	τ_{33}

Next, for a body in equilibrium, the stress components must satisfy the differential equations where X_1 , X_2 , X_3 are the components of the body force (per unit volume) acting on the body.

$$\frac{\partial \tau_{11}}{\partial x_1} + \frac{\partial \tau_{12}}{\partial x_2} + \frac{\partial \tau_{13}}{\partial x_3} + X_1 = 0,$$

$$\frac{\partial \tau_{21}}{\partial x_1} + \frac{\partial \tau_{22}}{\partial x_2} + \frac{\partial \tau_{23}}{\partial x_3} + X_2 = 0,$$

$$\frac{\partial \tau_{31}}{\partial x_1} + \frac{\partial \tau_{32}}{\partial x_2} + \frac{\partial \tau_{33}}{\partial x_3} + X_3 = 0,$$

Strain

- Deformation of a solid that can be related to stresses is described by strain.
- Take a string of an initial length L_0 .
- If it is stretched to a length L , it is natural to describe the change by dimensionless ratios such as L/L_0 , $(L - L_0)/L_0$, $(L - L_0)/L$.
- Use of dimensionless ratios eliminates the absolute length from consideration.
- The ratio L/L_0 is called the stretch ratio and is denoted by the symbol λ .
- The ratios are strain measures. Either of them can be used.

$$\varepsilon = \frac{L - L_0}{L_0}, \quad \varepsilon' = \frac{L - L_0}{L}$$

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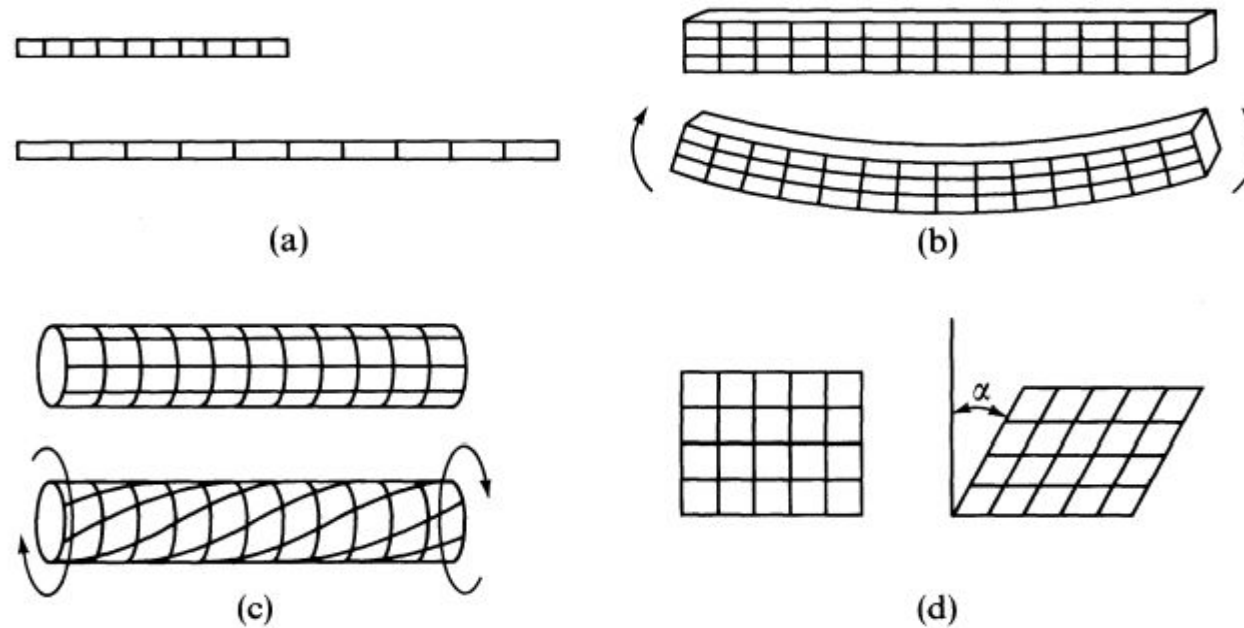


Figure 2.3:1 Patterns of deformation.

- The selection of proper strain measures is dictated basically by the stress-strain relationship (i.e., the constitutive equation of the material).
- For example, if we pull on a string, it elongates.
- The experimental results can be presented as a curve of the tensile stress σ plotted against the stretch ratio λ , or strain e .
- An empirical formula relating σ to e can be determined.
- The case of infinitesimal strain is simple because the different strain measures named above all coincide.
- It was found that for most engineering materials subjected to an infinitesimal strain in **uniaxial stretching**, a relation like

$$\sigma = Ee$$

(Contd..)

- is valid within a certain range of stresses, where E is a constant called Young's modulus.
- Equation is called Hooke's law. A material obeying Eq. is said to be a Hookean material.
- Steel is a Hookean material if (σ lies within certain bounds that are called yield stresses.
- Corresponding to Eq., the relationship for a Hookean material subjected to an infinitesimal shear strain is

$$\tau = G \tan \alpha,$$

- where G is another constant called the modulus of rigidity.

Viscoelasticity

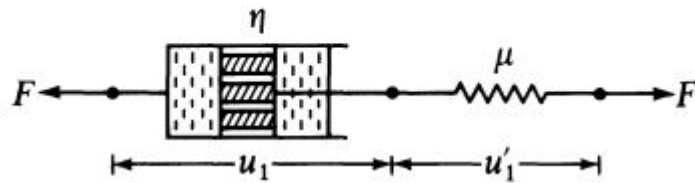
- When a body is suddenly strained and then the strain is maintained constant afterward, the corresponding stresses induced in the body decrease with time. This phenomena is called stress relaxation, or relaxation for short.
- If the body is suddenly stressed and then the stress is maintained constant afterward, the body continues to deform, and the phenomenon is called creep.
- If the body is subjected to a cyclic loading, the stress-strain relationship in the loading process is usually somewhat different from that in the unloading process, and the phenomenon is called hysteresis.
- The features of hysteresis, relaxation, and creep are found in many materials. Collectively, they are called features of viscoelasticity.

Models

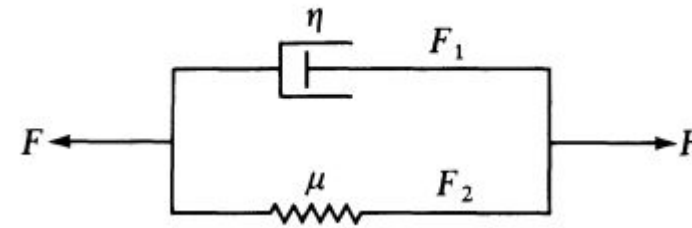
- Mechanical models are often used to discuss the viscoelastic behavior of materials.
- There are shown three mechanical models of material behavior, namely-
 - the Maxwell model
 - the Voigt model
 - the Kelvin model (also called the standard linear solid)

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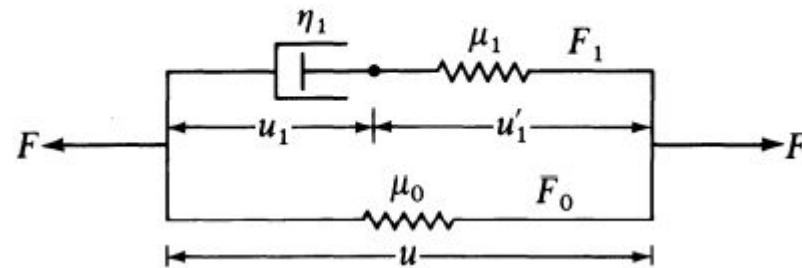
(a) A Maxwell body



(b) A Voigt body



(c) A Kelvin body (a standard linear solid)



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- All the models are composed of combinations of linear springs with spring constant μ and dash pots with coefficient of viscosity η .
- A linear spring is supposed to produce instantaneously a deformation proportional to the load.
- A dashpot is supposed to produce a **velocity proportional to the load at any instant**.
- Thus if F is the **force acting in a spring** and u is its extension, then $F = \mu u$.
- If the **force F acts on a dash pot**, it will produce a velocity of deflection U , and $F = \eta U$.

Maxwell model

- Now, in a Maxwell model, the same force is transmitted from the spring to the dashpot.
- This force produces a displacement F/μ in the spring and a velocity F/η in the dash pot.
- The velocity of the spring extension is $\dot{F}/\mu$ if we denote a differentiation with respect to time by a dot.
- The total velocity is the $\dot{u} = \frac{\dot{F}}{\mu} + \frac{F}{\eta}$ (Maxwell model).

Voigt Model

- For the Voigt model, the spring and the dashpot have the same displacement.
- If the displacement is u , the velocity is $\dot{u}$, and the spring and dash pot will produce forces μu and $\eta \dot{u}$, respectively.
- The total force F is therefore $F = \mu u + \eta \dot{u}$ (Voigt model).

Kelvin model

- For the Kelvin model (or standard linear model), let us break down the displacement u into u_1 of the dashpot and u_1' for the spring, whereas the total force F is the sum of the force F_0 from the spring and F_1 from the Maxwell element.
- Thus

$$(a) \quad u = u_1 + u_1',$$

$$(b) \quad F = F_0 + F_1,$$

$$(c) \quad F_0 = \mu_0 u,$$

$$(d) \quad F_1 = \eta_1 \dot{u}_1 = \mu_1 u_1'.$$

From this we can verify by substitution that

$$F = \mu_0 u + \mu_1 u_1' = (\mu_0 + \mu_1) u - \mu_1 u_1.$$

FLOW PROPERTIES OF BLOOD

Blood Rheology

- Blood contains many enzymes and hormones, and transports oxygen and carbon dioxide between the lungs and the cells of the tissues.
- Human blood is a suspension of cells in an aqueous solution of electrolytes and nonelectrolytes.
- By centrifugation, the blood is separated into plasma and cells.
- The plasma is about 90% water by weight, 7% plasma protein, 1 % inorganic substances, and 1 % other organic substances.
- The cellular contents are essentially all erythrocytes, or red cells, with white cells of various categories making up less than 1/600th of the total cellular volume, and platelets less than 1/800th of the cellular volume.

The Constitutive Equation of Blood Based on Viscometric Data and Casson's Equation

- Blood is a mixture of blood plasma and blood cells.
- In these viscometers, and, by implication, in large blood vessels whose diameters are much larger than the characteristic size of the blood cells, blood may be treated as a homogeneous fluid.
- The mechanical properties of the mixture as a homogeneous fluid can be described by a constitutive equation.

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- The constitutive equation of an isotropic incompressible Newtonian fluid is

$$\sigma_{ij} = -p\delta_{ij} + 2\mu V_{ij}, \quad (1)$$

where

$$V_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), \quad V_{ii} = V_{11} + V_{22} + V_{33} = 0. \quad (2)$$

- Here σ_{ij} is the stress tensor, V_{ij} is the strain-rate tensor, u_i is the velocity component, δ_{ij} is the isotropic tensor or Kronecker delta, p is the hydrostatic pressure, and μ is a constant called the coefficient of viscosity.

Other Aspects of Blood Rheology

- Actually the rheological properties of blood are more complex than what is portrayed above.
- If blood is tested dynamically, it can be made to reveal all features of viscoelasticity.
- Furthermore, the viscoelastic characteristics of blood change with the level of strain and strain history; hence it is **thixotropic**.
- These complex properties are probably not important in normal circulatory physiology, but can be significant when one tries to use blood rheology as a basis of clinical applications to diagnosis of diseases, pathology, or biochemical studies.

Laminar Flow of Blood in a Tube

- Let us consider the flow of blood in a circular cylindrical tube.
- We shall assume the flow to be laminar, that is, not turbulent.
- We shall also assume that the tube is long and the flow is steady, so that the conditions of flow change neither with the distance along the tube, nor with the time.

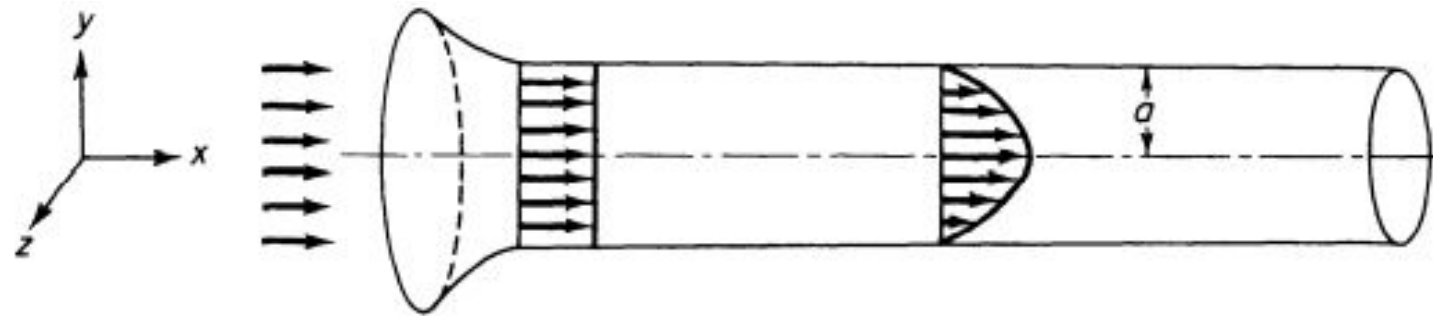


Figure 3.3:1 Velocity profiles in a steady laminar flow into a circular cylindrical tube.

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- We shall use polar coordinates for this problem.
- The polar axis coincides with the axis of the cylinder. See Fig. 3.3: 1.
- The flow obeys Navier-Stokes equations of motion of an incompressible fluid.
- The boundary condition is that blood adheres to the tube wall (the so-called no-slip condition).
- Since the boundary condition is axisymmetric, the flow is also axisymmetric and the only nonvanishing component of velocity is $u(r)$ in the axial direction; $u(r)$ is a function of r alone, and not of x .

(Contd..)

- Isolate a cylindrical body of fluid of radius r and unit length in the axial direction.
- This body is subjected to a pressure P_1 on the left-hand end, P_2 on the right-hand end, and shear stress τ on the circumferential surface. Since $P_1 - P_2 = -1 \cdot (dp/dx)$ acts on an area $2\pi r$, and τ acts on an area $1 \cdot \pi r^2$, we have, for equilibrium, the balance of forces

$$\tau \cdot 2\pi r = -\pi r^2 \frac{dp}{dx}, \quad \text{---(1)}$$

or

$$\tau = -\frac{r}{2} \frac{dp}{dx} \quad (\text{Stokes, 1851}).$$

A Newtonian Fluid

- By the definition of Newtonian viscosity, we have

$$\tau = -\mu \frac{du}{dr}. \quad \boxed{?} \quad 2$$

A substitution of Eq. (2) into Eq. (1) yields

$$\frac{du}{dr} = \frac{r}{2\mu} \frac{dp}{dx}.$$

A division of the rate of flow by the cross-sectional area of the tube yields the mean velocity of flow,

$$u_m = -\frac{a^2}{8\mu} \frac{dp}{dx}.$$

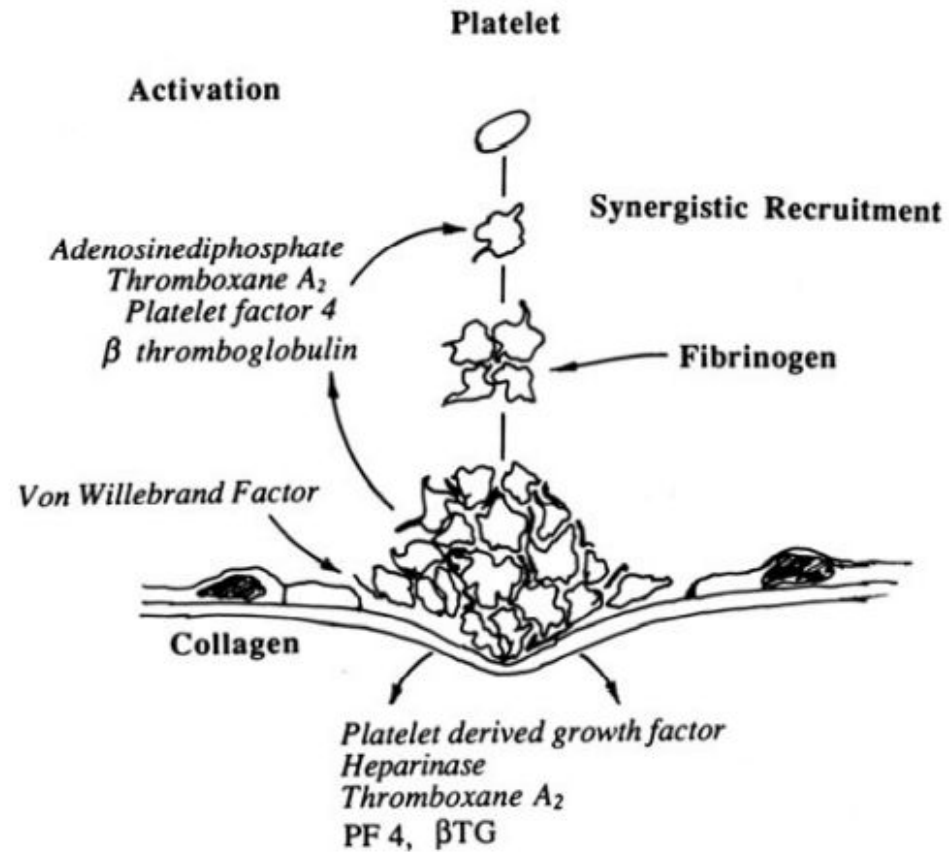
Thrombus Formation and Dissolution

- Blood clots are formed on an injured inner wall of blood vessels and on contact with the surfaces of medical devices.
- When a circulating blood comes into contact with such a surface, the platelets in the blood adhere to the surface, release a number of chemicals, attract more platelets to form a larger aggregation, generate thrombin, and form fibrin, resulting in a thrombus.

Thrombogenesis

- The injured endothelium exposes collagen of the basement membrane which interacts with the glycoprotein on the platelet membrane and the **von Willebrand factor** (a blood glycoprotein that promotes hemostasis, specifically, platelet adhesion) which is synthesized by endothelial cells and is present in the plasma, and on the platelets.
- The adherent platelets then release adenosine diphosphate (ADP), thromboxane A₂, fibrinogen, factor V, platelet factor 4, beta thromboglobulin, and a platelet-derived growth factor.

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- The released ADP and thromboxane A₂ act synergistically to recruit circulating platelets and enlarge the aggregate.
- Essential to this step is the mobilization of a platelet-membrane fibrinogen receptor and the calcium dependent interplatelet bridging by fibrinogen.

BIOVISCOELASTIC SOLIDS

Actin

- Actin molecules are present in all muscles, leukocytes, red blood cells, endothelial cells, and many other cells.
- The measurement is based on the fact that a single actin filament (7 nm in diameter) can be clearly seen by video-fluorescence microscopy.
- Actin monomers are globular.
- They polymerize into filaments.

Elastin

- Elastin is the most "linearly" elastic biosolid materials known.
- Elastin is a protein found in vertebrates.
- It is present as thin strands in the skin and in areolar connective tissue.
- It forms quite a large proportion of the material in the walls of arteries and veins, especially near the heart.
- It is a prominent component of the lung tissue.

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- The ligamentum nuchae, which runs along the top of the neck of horses and cattle, is almost pure elastin.
- These ligaments also contain a small amount of collagen, which can be denatured by heating to 66°C or above.
- Heating to this degree and cooling again does not change the mechanical properties of elastin.

The Elastin Molecule

- The molecular structure of the tropoelastin, a precursor molecule of elastin, has been sequenced.
- Mecham and Heuser (1991) have shown that tropoelastin is formed intracellularly and then crosslinked extracellularly.
- The mature, cross-linked elastin molecule is inert and so stable that in normal circumstances it lasts in the body for the entire life of the organism.
- Repeating sequences in elastin molecule have been noted, and some of their analogs have been prepared chemically, and studied thermo-mechanically.
- Of these, poly (V PG VG), poly (V PG F GV GAG), and poly (VPGG) on γ -irradiation cross-linking have been shown to be elastic.

Resilin and Abductin

- A biosolid similar to elastin in mechanical behavior but quite different in chemical composition is resilin.
- Resilin is a **protein found in arthropods**.
- It is hard when it is dry, but in the natural state it contains 50% to 60% water and is soft and rubbery.
- Dried resilin can be made rubbery again by soaking it in water.
- It can be stretched to three times its initial length.

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- An elastic protein found in scallops' hinges is abductin.
- Scallops use it to open the valves (the adductor muscle is used to close the valves).
- The elastic moduli of abductin and elastin are about the same.

Fibers

- Fibers form a distinctive class of polymeric materials.
- X-ray diffraction shows that fibers contain both crystalline regions, where the molecules are arranged in orderly patterns, and amorphous regions, where they are arranged randomly (see Hearle, 1963).
- A typical example is collagen bundle. Plant fibers and synthetic textile fibers belong to this class.

Collagen

- Collagen is a basic structural element for soft and hard tissues in animals.
- It gives mechanical integrity and strength to our bodies.
- It is present in a variety of structural forms in different tissues and organs.
- Its importance to man may be compared to the importance of steel to our civilization: steel is what most of our vehicles, utensils, buildings, bridges, and instruments are made of.
- Collagen is the main load carrying element in blood vessels, skin, tendons, cornea, sclera, bone, fascia, dura mater, the uterian cervix, etc.

Quasi-Linear Viscoelasticity of Soft Tissues

- The experimental results illustrates that biological tissues are not elastic.
- The history of strain affects the stress.
- In particular, there is a considerable difference in stress response to loading and unloading.
- Most authors discuss soft tissue experiments in the framework of the linear theory of viscoelasticity relating stress and strain on the basis of the Voigt, Maxwell, and Kelvin models.

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- Let us consider a cylindrical specimen subjected to tensile load.
- If a step increase in elongation (from $\lambda = 1$ to λ) is imposed on the specimen, the stress developed will be a function of time as well as of the stretch λ .
- The history of the stress response, called the relaxation function, and denoted by $K(\lambda, t)$, is assumed to be of the form

$$K(\lambda, t) = G(t)T^{(e)}(\lambda), \quad G(0) = 1, \quad (1)$$

(Contd..)

- in which $G(t)$, a normalized function of time, is called the reduced relaxation function, and $T^{(e)}(\lambda)$, a function of λ alone, is called the elastic response.
- We then assume that the stress response to an infinitesimal change in stretch, $\delta\lambda(\tau)$, superposed on a specimen in a state of stretch λ at an instant of time τ is, for $t > \tau$:

$$G(t - \tau) \frac{\partial T^{(e)}[\lambda(\tau)]}{\partial \lambda} \delta\lambda(\tau). \quad (2)$$

- Finally, we assume that the superposition principle applies, so that

$$T(t) = \int_{-\infty}^t G(t - \tau) \frac{\partial T^{(e)}[\lambda(\tau)]}{\partial \lambda} \frac{\partial \lambda(\tau)}{\partial \tau} d\tau, \quad (3)$$

(Contd..)

- that is, the tensile stress at time t is the sum of contributions of all the past changes, each governed by the same reduced relaxation function.
- Rewriting Eq. (3) in the form

$$T(t) = \int_{-\infty}^t G(t - \tau) \dot{T}^{(e)}(\tau) d\tau, \quad (4)$$

- where a dot denotes the rate of change with time, we see that the stress response is described by a linear law relating the stress T with the elastic response $\dot{T}^{(e)}$.

Anisotropy

- Bone is said to be anisotropic i.e., material properties are different in different directions.
- Bone is composed of mainly collagen and other minerals (hydroxyapatite or calcium).
- Collagen has high tensile strength and low compressive strength.
- Hydroxyapatite is stiff, brittle and has high compressive strength.
- Anisotropic material can resist different types of forces and is strongest in compression, weakest in shear and intermediate in tension.

Types of Fractures



A *greenstick* fracture is incomplete, and the break occurs on the convex surface of the bend in the bone.



A *fissured* fracture involves an incomplete longitudinal break.



A *comminuted* fracture is complete and fragments the bone.



A *transverse* fracture is complete, and the break occurs at a right angle to the axis of the bone.



An *oblique* fracture occurs at an angle other than a right angle to the axis of the bone.



A *spiral* fracture is caused by twisting a bone excessively.